High Frequency Words in Reuters Financial News

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DATA 620 – Web Analytics

Introduction

This assignment looks at word frequencies in one corpus and asks how they compare with the "typical" frequencies of English. I chose the **Reuters-21578 news corpus**, the version that ships with NLTK. It's a collection of business and financial newswire stories from Reuters, written in 1987 and tagged with 90 topic categories (earnings, acquisitions, crude oil, grain, money supply, trade and so on).

I picked it because it's the kind of text a data scientist working in finance would actually mine, and because financial news has its own shorthand: "mln," "dlrs," "pct," "shr." That shorthand makes the last question, how this corpus differs from all words in all corpora, worth answering. To stand in for "all corpora" I use the **Brown corpus**, also in NLTK: about a million words of American English drawn from 15 genres, from news to romance fiction.

The questions from the assignment:

1. Choose a corpus of interest.
2. How many total unique words are in the corpus?
3. Taking the most common words, how many unique words represent half of the total words?
4. Identify the 200 highest frequency words.
5. Create a graph that shows the relative frequency of these 200 words.
6. Does the observed relative frequency of these words follow Zipf's law?
7. In what ways might the frequency of words in this corpus differ from "all words in all corpora"?

1. The corpus

NLTK's copy of Reuters-21578 is the "ApteMod" split, which keeps the stories that have at least one topic label. Each story starts with its headline in capital letters, followed by the body.

The corpus has **10,788 stories** in **90 categories**. The largest categories are shown below (a story can have more than one).

Category	Stories
earn	3964
acq	2369
money-fx	717
grain	582

Category	Stories
crude	578
trade	485
interest	478
ship	286

```
LSB INDUSTRIES INC &lt;LSB> 4TH QTR NET
  Shr profit five cts vs loss 2.11 dlrs
      Net profit 375,061 vs loss 10.4 mln
      Revs 39.9 mln vs 37.8 mln
      Avg shrs 6,536,008 vs 4,939,895
      12 mths
      Shr profit 47 cts vs loss 3.37 dlrs
      Net profit 2,837,288 vs loss 16.6 mln
      Revs 169.1 mln vs 149.4 mln
      Avg shrs 6,037,640 vs 4,937,378
      NOTE: primary earnings per share are based on the weighted
  average number of common and dilutive common equivalent shares
  outstanding during each period after accounting for preferred
  stock dividends.
  [… 855 more characters of notes on restructuring charges and taxes …]
```

This excerpt (the start of one earnings story) shows why this corpus needs care before counting. Stories mix capitalized headlines with normal text, abbreviations such as "U.S." have periods inside them, company tickers appear inside HTML-escaped brackets (`<ABC>`), and earnings reports are written as compact tables such as "Shr 22 cts vs 18 cts." NLTK's built-in `reuters.words()` splits "U.S." into "U", ".", "S", "." and turns `<` into a word "lt," so I tokenize the raw text myself.

Tokenizing rules

- Decode the HTML entities and lowercase everything, so "OIL" in a headline and "oil" in the body count as the same word.
- A word is a run of letters, optionally joined by a period, apostrophe, ampersand or hyphen (so "u.s", "at&t" and "mid-year" each stay one word).
- A possessive "'s" is removed, so "Japan's" counts as "japan".
- Numbers are not words. They're counted separately because they're a big part of this corpus.

After tokenizing, the corpus has **1,292,829 words** (plus 95,177 numbers, which are left out). Numbers make up 7% of all word and number tokens, a sign of how numeric financial news is.

2. How many unique words?

"Unique word" can be defined in several ways, and the answer changes a lot depending on which you pick. The table builds up from the loosest definition to the strictest:

- **NLTK raw tokens:** every distinct token from `reuters.words()`, punctuation and numbers included, with case kept ("Oil" and "oil" are different).
- **Words, case kept:** my tokenizer above, but without lowercasing.
- **Words, lowercased:** my main definition.

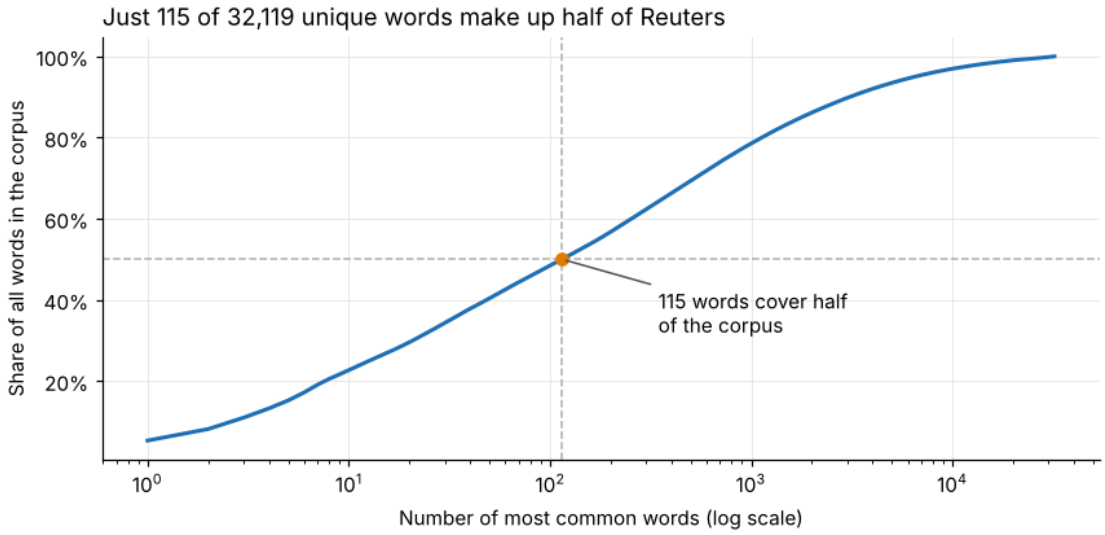
- **Lowercased, stopwords removed:** drops NLTK's 179 English stopwords ("the," "of," "and" …).
- **Lemmatized:** groups inflected forms with WordNet ("shares" → "share," "rates" → "rate").
- **Used more than once:** the lowercased words that appear at least twice, which drops one-off typos, rare names and ticker codes.

Definition of a unique word	Unique words
NLTK raw tokens (incl. punctuation and numbers)	41,600
Words, case kept	42,831
Words, lowercased (main definition)	32,119
Lowercased, stopwords removed	31,954
Lowercased, stopwords removed, lemmatized	29,556
Lowercased, used more than once	19,447

Answer: using lowercased words as the definition, the corpus has **32,119 unique words** out of 1,292,829 total words. Lowercasing alone removes 10,712 duplicates, mostly words that appear in both capitalized headlines and body text. **12,672 words (39% of the vocabulary) appear only once**, and many of those are ticker codes such as "bti.l" or "mc.t" and company names. That long tail of rare words is typical of natural language, and it comes back in the Zipf section. Removing stopwords changes the count by only 165, because the stopword list is short, even though those few words make up a large share of the text (section 7).

3. How many words make up half the corpus?

Sort the unique words from most to least common and add up their counts until they reach half of all words in the corpus.



Answer: 115 unique words (0.36% of the vocabulary) account for half of the 1,292,829 words

in the corpus. Most of them are function words such as “the,” “of” and “to,” but the list already includes Reuters shorthand like “mln,” “dlrs” and “pct.” If stopwords are removed first, it takes 317 words to cover half of what remains, because content words are spread over a much larger vocabulary.

4. The 200 most frequent words

The table lists the top 200 words with their count and relative frequency (count divided by all words in the corpus), in two tables of 100 (ranks 1–100, then 101–200), each split into two side-by-side blocks of 50.

Rank	Word	Count	Share of all words	Rank	Word	Count	Share of all words
1	the	69247	5.36%	51	were	3092	0.24%
2	of	36749	2.84%	52	trade	3064	0.24%
3	to	36280	2.81%	53	had	2975	0.23%
4	in	29217	2.26%	54	profit	2939	0.23%
5	and	25629	1.98%	55	about	2812	0.22%
6	said	25382	1.96%	56	market	2790	0.22%
7	a	24841	1.92%	57	new	2707	0.21%
8	mln	18608	1.44%	58	qtr	2672	0.21%
9	vs	14341	1.11%	59	shares	2652	0.21%
10	for	13465	1.04%	60	stock	2625	0.20%
11	dlrs	12414	0.96%	61	one	2539	0.20%
12	it	11100	0.86%	62	also	2532	0.20%
13	pct	9784	0.76%	63	they	2528	0.20%
14	on	9094	0.70%	64	tonnes	2510	0.19%
15	cts	8361	0.65%	65	revs	2312	0.18%
16	from	8217	0.64%	66	two	2278	0.18%
17	is	7668	0.59%	67	sales	2215	0.17%
18	that	7540	0.58%	68	prices	2194	0.17%
19	its	7402	0.57%	69	up	2191	0.17%
20	by	7082	0.55%	70	group	2112	0.16%
21	at	7015	0.54%	71	may	2110	0.16%
22	net	6987	0.54%	72	been	2070	0.16%
23	year	6771	0.52%	73	per	2058	0.16%
24	be	6354	0.49%	74	march	2030	0.16%
25	with	6179	0.48%	75	april	2011	0.16%
26	will	5951	0.46%	76	after	1958	0.15%
27	was	5816	0.45%	77	more	1881	0.15%
28	billion	5811	0.45%	78	first	1881	0.15%
29	u.s	5656	0.44%	79	japan	1879	0.15%
30	he	5210	0.40%	80	rate	1845	0.14%
31	loss	5115	0.40%	81	co	1840	0.14%
32	has	4864	0.38%	82	price	1794	0.14%
33	would	4671	0.36%	83	quarter	1788	0.14%
34	company	4661	0.36%	84	than	1771	0.14%
35	an	4556	0.35%	85	february	1771	0.14%

Rank	Word	Count	Share of all words	Rank	Word	Count	Share of all words
36	as	4526	0.35%	86	offer	1717	0.13%
37	not	4386	0.34%	87	government	1706	0.13%
38	shr	4180	0.32%	88	january	1689	0.13%
39	inc	4121	0.32%	89	other	1653	0.13%
40	which	3666	0.28%	90	dollar	1627	0.13%
41	bank	3625	0.28%	91	week	1601	0.12%
42	but	3601	0.28%	92	exchange	1599	0.12%
43	this	3516	0.27%	93	note	1584	0.12%
44	corp	3397	0.26%	94	if	1555	0.12%
45	last	3228	0.25%	95	agreement	1550	0.12%
46	or	3224	0.25%	96	against	1492	0.12%
47	are	3215	0.25%	97	no	1490	0.12%
48	oil	3205	0.25%	98	could	1468	0.11%
49	have	3107	0.24%	99	production	1456	0.11%
50	share	3102	0.24%	100	today	1445	0.11%

Rank	Word	Count	Share of all words	Rank	Word	Count	Share of all words
101	we	1441	0.11%	151	months	1022	0.08%
102	ltd	1440	0.11%	152	into	1020	0.08%
103	told	1427	0.11%	153	end	1017	0.08%
104	foreign	1425	0.11%	154	money	1015	0.08%
105	some	1422	0.11%	155	banks	1008	0.08%
106	international	1420	0.11%	156	major	1000	0.08%
107	over	1410	0.11%	157	department	983	0.08%
108	three	1374	0.11%	158	ec	976	0.08%
109	expected	1373	0.11%	159	cash	975	0.08%
110	their	1370	0.11%	160	american	970	0.08%
111	rose	1366	0.11%	161	officials	969	0.07%
112	interest	1361	0.11%	162	spokesman	969	0.07%
113	rise	1322	0.10%	163	report	962	0.07%
114	rates	1319	0.10%	164	japanese	957	0.07%
115	total	1318	0.10%	165	made	956	0.07%
116	under	1307	0.10%	166	now	953	0.07%
117	current	1306	0.10%	167	investment	951	0.07%
118	record	1293	0.10%	168	west	948	0.07%
119	there	1282	0.10%	169	imports	946	0.07%
120	dlr	1279	0.10%	170	buy	946	0.07%
121	all	1265	0.10%	171	mths	946	0.07%
122	nine	1263	0.10%	172	countries	944	0.07%
123	stg	1224	0.09%	173	analysts	942	0.07%
124	tax	1216	0.09%	174	december	938	0.07%
125	exports	1210	0.09%	175	acquisition	938	0.07%
126	five	1196	0.09%	176	while	935	0.07%

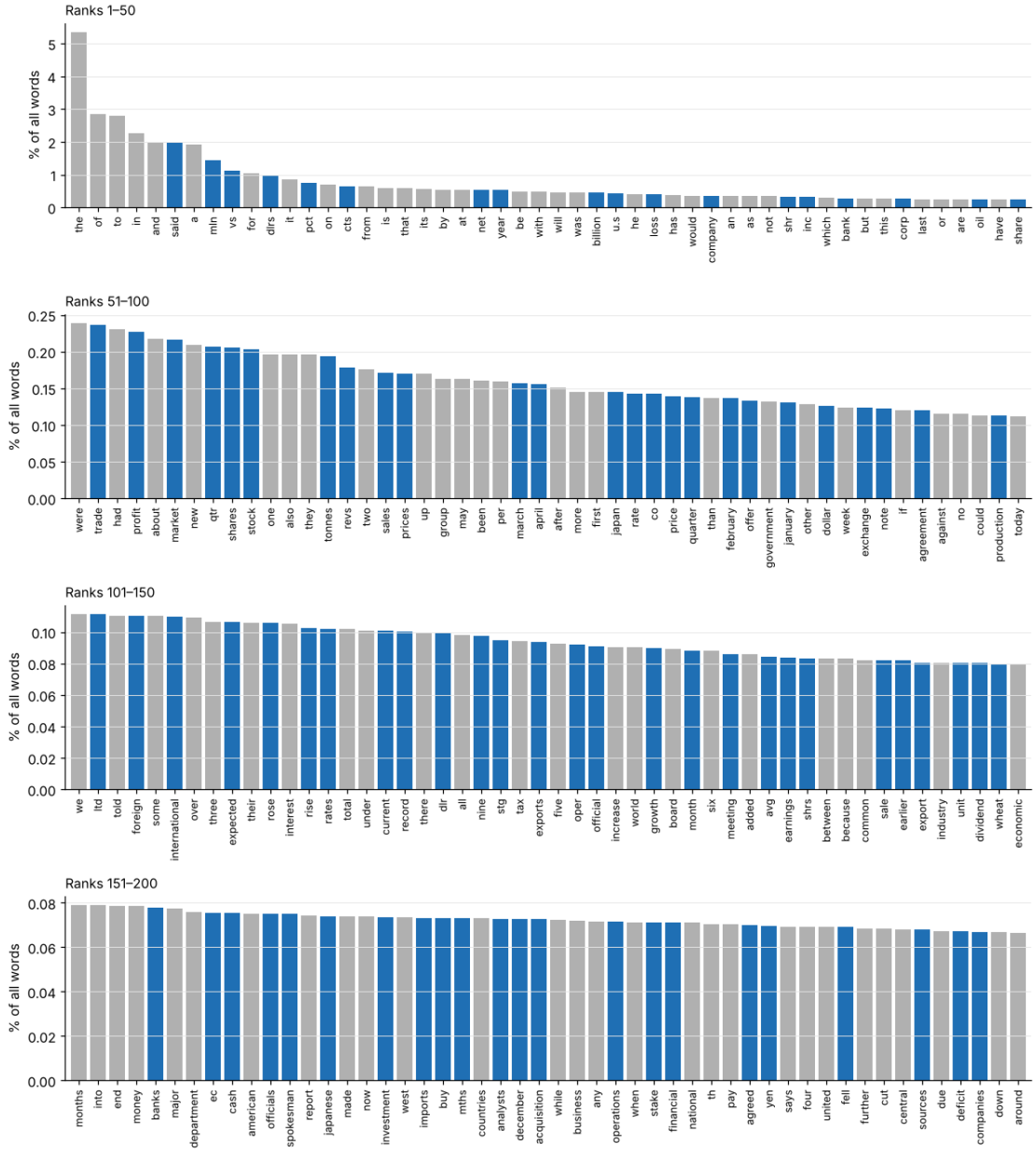
Rank	Word	Count	Share of all words	Rank	Word	Count	Share of all words
127	oper	1192	0.09%	177	business	932	0.07%
128	official	1176	0.09%	178	any	925	0.07%
129	increase	1171	0.09%	179	operations	923	0.07%
130	world	1169	0.09%	180	when	921	0.07%
131	growth	1159	0.09%	181	stake	920	0.07%
132	board	1152	0.09%	182	financial	919	0.07%
133	month	1137	0.09%	183	national	919	0.07%
134	six	1137	0.09%	184	th	912	0.07%
135	meeting	1112	0.09%	185	pay	909	0.07%
136	added	1108	0.09%	186	agreed	907	0.07%
137	avg	1086	0.08%	187	yen	898	0.07%
138	earnings	1082	0.08%	188	says	895	0.07%
139	shrs	1078	0.08%	189	four	895	0.07%
140	between	1076	0.08%	190	united	895	0.07%
141	because	1076	0.08%	191	fell	894	0.07%
142	common	1064	0.08%	192	further	887	0.07%
143	sale	1062	0.08%	193	cut	882	0.07%
144	earlier	1059	0.08%	194	central	881	0.07%
145	export	1041	0.08%	195	sources	877	0.07%
146	industry	1040	0.08%	196	due	871	0.07%
147	unit	1039	0.08%	197	deficit	868	0.07%
148	dividend	1037	0.08%	198	companies	864	0.07%
149	wheat	1026	0.08%	199	down	863	0.07%
150	economic	1022	0.08%	200	around	859	0.07%

Together, the top 200 words make up **57% of all words** in the corpus. 59 of them are NLTK stopwords. The rest are dominated by finance and business vocabulary: units ("mln," "billion," "dlrs," "pct," "cts"), earnings-table words ("shr," "net," "vs," "qtr," "revs"), company words ("inc," "corp," "company," "shares") and commodities and markets ("oil," "bank," "trade," "market," "prices").

5. Relative frequency of the top 200 words

Each bar is one word's share of all words in the corpus, in rank order. To set up question 7, words are colored **blue when they're at least five times more common in Reuters than in the Brown corpus** of general English, and grey otherwise.

After a steep drop, the top 200 Reuters words flatten out; 92 of them (blue) are at least 5x more common than in general English

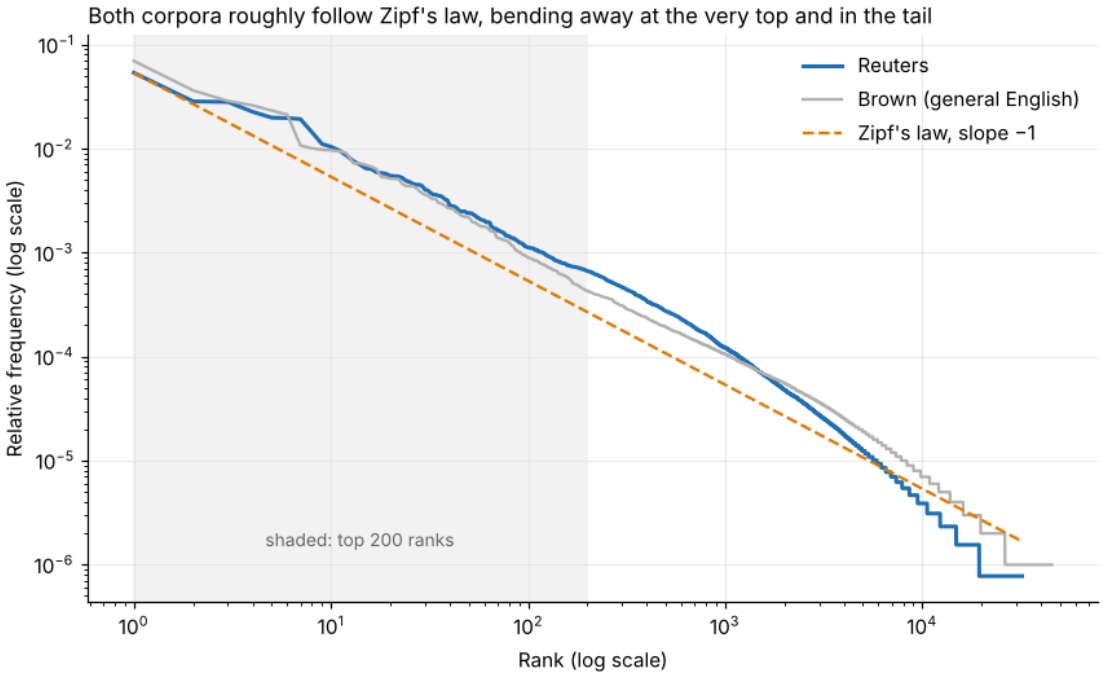


The first panel shows the classic shape: “the” alone is 5.4% of all words, and by rank 50 a word is down to about 0.24%. Each panel has its own y-axis, because on a shared axis the last three panels would be nearly flat. **92 of the top 200 words** are blue. They show up at every rank, from “mln” and “dlrs” near the top to words like “tonnes” and “dividend” further down.

6. Does the corpus follow Zipf's law?

Zipf's law says a word's frequency is roughly inversely proportional to its rank: the 2nd most common word appears about half as often as the 1st, the 10th about a tenth as often, and so on. In symbols, $f(r) \approx C/r^s$ with $s \approx 1$. On a log-log plot of frequency against rank that's a straight line with slope -1.

I check this three ways: plot rank against frequency on log-log axes with a slope -1 reference line, fit the slope by least squares, and check whether rank $\times$ frequency stays roughly constant.



rank	word	count	rel_freq	rank × freq	Zipf predicts
1	the	69247	0.0536	0.054	0.0536
2	of	36749	0.0284	0.057	0.0268
3	to	36280	0.0281	0.084	0.0179
5	and	25629	0.0198	0.099	0.0107
10	for	13465	0.0104	0.104	0.0054
20	by	7082	0.0055	0.11	0.0027
50	share	3102	0.0024	0.12	0.0011
100	today	1445	0.0011	0.112	0.0005
150	economic	1022	0.0008	0.119	0.0004
200	around	859	0.0007	0.133	0.0003

Answer: roughly yes, with the usual departures. The fitted log-log slope for the top 200 words is **-0.93**, against Zipf's ideal of -1 (Brown's top 200 give -1.04). Over ranks 10 to 5,000 the slope is -1.22. So across the top 200, frequency falls off with rank at close to the rate Zipf's

law predicts. On the chart, the blue line sits above the orange reference (because the very top is flatter, see below), then falls a little faster than it and meets it near rank 10,000. Over four orders of magnitude the line stays within a factor of about 2.7 of the reference.

The fit isn't perfect, in three places:

- **The very top is flatter than Zipf predicts.** "of" and "to" (ranks 2 and 3) appear more often than 1/2 and 1/3 of "the," so rank × frequency rises from 0.054 at rank 1 to about 0.10–0.13 from rank 10 on, instead of staying constant. This is common in English.
- **The middle of the top 200 has bumps.** Words from earnings tables ("vs," "cts," "shr," "net") are pushed up the ranking because thousands of short earnings reports repeat the same template. A corpus made of a single genre bends the curve this way.
- **The tail drops off faster than the line.** Past a few thousand words the curve falls more steeply, and the last 12,672 words are a flat step of words used only once. That's partly a sample-size effect: a corpus of 1.3 million words can't show the rare words a larger sample would contain.

7. How does this corpus differ from "all words in all corpora"?

There's no true "all corpora," so I use the Brown corpus as the stand-in for general English. It was also compiled from American text (published in 1961) and covers 15 genres, from press reports to fiction. Comparing relative frequencies, word by word, shows what is special about financial news.

Words much more common in Reuters than in general English

Word	Reuters per 10k	Brown per 10k	Ratio
mln	143.9	0	14,470x
dlrs	96	0	9,654x
pct	75.7	0	7,609x
cts	64.7	0	6,502x
shr	32.3	0	3,251x
qtr	20.7	0	2,078x
tonnes	19.4	0	1,952x
revs	17.9	0	1,799x
dlr	9.9	0	995x
stg	9.5	0	953x
oper	9.2	0	928x
avg	8.4	0	845x
shrs	8.3	0	839x
mths	7.3	0	736x
div	5.3	0	534x

Words much less common in Reuters than in general English

Word	Reuters per 10k	Brown per 10k	Ratio
her	0.1	30.2	1/355
boy	0	2.6	1/333

Word	Reuters per 10k	Brown per 10k	Ratio
mother	0	2.5	1/325
woman	0	2.4	1/318
girl	0	2.3	1/297
god	0	3.5	1/228
husband	0	1.5	1/192
girls	0	1.4	1/186
eyes	0	4	1/172
beautiful	0	1.3	1/165
religion	0	1.2	1/154
gun	0	1.2	1/154
church	0	3.5	1/153
doctor	0	1.2	1/153
poet	0	1.1	1/145

Measure	Reuters	Brown
Total words	1,292,829	1,005,266
Unique words (lowercased)	32,119	45,487
Words covering half the corpus	115	125
Top-200 words shared by both corpora	76 of 200	76 of 200
Share of words that are stopwords	34%	48%
Share of words that are personal pronouns (I, me, my, we, you, he, him, his, she, her)	0.7%	3.9%

Answer. The comparison shows five main differences:

1. **A specialized vocabulary takes over the top ranks.** Only 76 of the top 200 Reuters words are also in Brown's top 200. Words such as "mln," "dlrs," "pct," "cts" and "shr" are among the most common words in Reuters but almost never appear in general English. These are Reuters' house abbreviations for million, dollars, percent, cents and share.
2. **Templates repeat.** Thousands of earnings reports follow the same layout ("Shr 22 cts vs 18 cts … Net … Revs …"), so a handful of template words get far more weight than they would in varied writing. It takes 115 words to cover half of Reuters against 125 for Brown, even though Reuters is the larger corpus, and Reuters has fewer unique words (32,119 vs 45,487). Both point the same way: financial news reuses a smaller vocabulary.
3. **Fewer personal pronouns.** Personal pronouns are 0.7% of Reuters but 3.9% of Brown. Newswire reports what organizations and officials "said" (the 6th most common word); it has no characters or narrators, so "I," "you," "she" and "her" are rare. The most under-represented words are about people and private life: "her," "mother," "girl," "husband," "eyes," "church."
4. **Numbers, place names and companies.** Numbers make up 7% of tokens, and words like "u.s," "japan," "inc" and "corp" are far more common than in general text. The long tail is full of ticker codes, which general English wouldn't have at all.
5. **The time period.** The stories are from 1987, so the vocabulary reflects that year's news: OPEC and crude oil, the U.S.–Japan trade dispute, and Bundesbank and Federal Reserve

policy. A financial corpus from today would have different topical words (crypto, inflation, AI) even though the template words would probably remain.

The very top of the list is shared: "the," "of," "to," "and," "a" and "in" are the six most common words in Brown and among the seven most common in Reuters. That part of the distribution seems close to universal for English. Below it the corpora split apart. Stopwords are only 34% of Reuters against 48% of Brown, because so much of a financial story is units, abbreviations, company names and numbers rather than connecting words.

Summary

Question	Answer
Corpus	Reuters-21578 (NLTK ApteMod): 10,788 financial news stories from 1987
Unique words	32,119 lowercased words; between 19,447 and 42,831 depending on the definition
Words covering half the corpus	115 words cover half of the 1,292,829 words
Top 200 words	57% of all words; 59 are stopwords, most of the rest are finance and business terms
Zipf's law	Roughly followed: slope -0.93 for the top 200; flatter at the very top, steeper in the tail
Difference from general English	Specialized abbreviations, repeated earnings templates, few personal pronouns, many numbers and company names, 1987 topics

Limitations. The results depend on the tokenizing choices (for example, keeping "u.s" as one word and dropping numbers). Brown is only a proxy for "all corpora" and is itself 1961 American English. The corpus covers only stories that were given topic labels, and earnings and acquisition stories make up most of it, so the results describe that mix of financial news rather than financial writing in general.